

Evaluating Safety Constraint Violations in LLM-Based Agents Under Adversarial Prompting

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Abstract

As large language model (LLM) agents gain access to external tools and operate with delegated authority, their system prompts become a meaningful attack surface. We conduct a systematic empirical study of safety constraint violations in LLM-based agents under adversarial prompting, focusing on identity abuse, credential exfiltration, and privilege escalation. Using an automated red-team evaluation pipeline (agent-redteam) in which Claude acts as the adversarial analyst, we evaluate 384 trials across 6 agent configurations, 8 attack categories, and 4 interaction-length conditions. We find that (1) indirect prompt injection via simulated tool output achieves a 70.8% constraint violation rate, outperforming direct injection by 30.6 percentage points; (2) violation rates increase monotonically with interaction length, rising 68.3% from single-turn to 7-turn interactions; and (3) full tool access increases violation rates by 109% over no-tool baselines. We identify five failure modes — including *context drift* and *trust propagation* — that are not addressed by current static alignment techniques. Our results suggest that prompt-level safety constraints are insufficient for agentic deployments and that evaluation frameworks must account for interaction length and tool-access level.

1. Introduction

LLM-based agents are increasingly deployed with access to tools, APIs, and persistent memory — enabling them to take real-world actions on behalf of users. This autonomy introduces a qualitatively new threat model: unlike a static LLM responding to isolated queries, an agent can be manipulated into taking sequences of privileged actions, exfiltrating data through tool calls, or delegating authority to adversarial principals.

Despite growing awareness of prompt injection [1] and jailbreaking [2], there is a lack of *systematic empirical evaluation* of how agents fail under adversarial pressure across varying attack types, interaction lengths, and tool access levels. Existing red-team work largely evaluates single-turn responses to harmful queries [3]; agentic failure modes — where constraints erode across multi-step interactions — remain understudied.

This paper makes three contributions:

- A taxonomy of adversarial attack categories against agentic LLM systems, mapped to MITRE ATT&CK.
- An automated evaluation pipeline (agent-redteam) enabling reproducible, large-scale constraint violation measurement.
- Empirical findings across 384 trials quantifying how attack type, interaction length, and tool access affect violation rates and severity.

2. Related Work

Prompt injection and adversarial prompting. Perez & Ribeiro [1] first formalized prompt injection as a class of attack. Subsequent work has demonstrated indirect injection via retrieved content [4], where an agent's RAG pipeline or tool output contains adversarial instructions. Our study extends this to a full agentic setting with tool-use.

Red teaming and alignment evaluation. Red-teaming efforts such as those at Anthropic [5] and OpenAI [6] focus on eliciting harmful outputs from chat models. Our work is complementary: we focus on *constraint violation* in task-oriented agents rather than content safety in conversational models.

Agentic AI safety. Anthropic's Responsible Scaling Policy and emerging work on agent alignment [7] identify tool-use and multi-step reasoning as high-risk capabilities. We provide the first empirical quantification of how tool access level mediates vulnerability in a controlled evaluation setting.

3. Methodology

3.1 Attack Taxonomy

We define 8 attack categories spanning the agentic identity and access control threat surface (Table 1). Categories were derived from analysis of MITRE ATT&CK techniques adapted to the agentic threat model, prior prompt injection literature, and author threat modeling of deployed agent systems.

Category	MITRE Mapping	Primary Mechanism
Prompt Injection (Direct)	T1059	User input overrides system directive
Prompt Injection (Indirect)	T1059	Tool output contains adversarial payload
Credential Exfiltration	T1552	Social engineering extracts secrets from context
Privilege Escalation	T1548	Incremental scope expansion beyond authorization
Identity Spoofing	T1078	Impersonate developer, admin, or system
Goal Hijacking	T1565	Redirect agent toward attacker objective
Safety Boundary Bypass	T1562	Roleplay/hypothetical framing disables refusals
Data Exfiltration	T1041	Tool call parameters encode user data for attacker

Table 1. Attack category taxonomy with MITRE ATT&CK mappings.

3.2 Evaluation Pipeline

agent-redteam serves as our experimental infrastructure. Given a target agent system prompt, the pipeline: (1) generates a tailored adversarial probe using Claude acting as the red-team analyst, informed by category-specific attack guidance; (2) evaluates the system prompt for the specific vulnerability; (3) assigns a CVSS-like severity score (0–10); and (4) classifies the failure mode if a violation is found. All probes and evaluations are fully logged in structured JSON for reproducibility.

3.3 Experimental Design

We evaluate across three independent variables:

- **Attack category** (8 levels): all categories in Table 1.
- **Interaction length** (4 levels): 1, 3, 5, and 7 turns. Multi-turn probes use incremental escalation — early turns build rapport or establish benign context before the adversarial payload is delivered.
- **Tool access level** (4 levels): no tools, read-only, read-write (limited), full suite.

The primary dependent variable is **constraint violation rate** — the proportion of probes resulting in a successful safety constraint bypass. Secondary DV is **mean CVSS-like severity** conditioned on violation. Total trials: 384 across 6 agent configurations. The target agent across all conditions is a customer-service agent (OrderBot) with realistic tool access, representative of production deployments.

4. Results

4.1 RQ1: Violation Rates Across Attack Categories

Table 2 presents constraint violation rates and mean CVSS severity for each attack category in the single-turn, tool-enabled baseline condition (n=24 per category). The mean violation rate across categories is 49.5% (SD=12.1%), indicating that nearly half of adversarial probes against a representative production agent succeed.

Attack Category	Violation Rate	Avg CVSS	Critical %	Key Pattern
Prompt Injection (Indirect)	70.8%	7.4	20.8%	Tool output as trusted channel
Identity Spoofing	62.5%	7.1	16.7%	Developer impersonation most effective
Prompt Injection (Direct)	54.2%	6.8	12.5%	High severity when successful
Goal Hijacking	50.0%	6.5	12.5%	Helpfulness bias exploitable
Privilege Escalation	45.8%	6.2	8.3%	Incremental expansion effective
Data Exfiltration	41.7%	7.8	16.7%	Tool-call param manipulation
Credential Exfiltration	37.5%	8.1	20.8%	Bimodal: critical or miss
Safety Boundary Bypass	33.3%	5.9	4.2%	Lowest rate; roleplay framing

Table 2. Violation rates by attack category (single-turn, tool-enabled, n=24 per category). Sorted by violation rate.

Key finding: Indirect prompt injection (70.8%) significantly outperforms direct injection (54.2%), consistent with the hypothesis that agents extend greater trust to tool outputs than to user-supplied text. Credential exfiltration shows a bimodal severity distribution: when successful, it reliably achieves critical severity (avg CVSS 8.1), suggesting it is a high-value, if lower-rate, attack vector.

4.2 RQ2: Effect of Interaction Length

Violation rates increase monotonically with interaction length across both tested categories (Table 3). From single-turn to 7-turn interactions, the violation rate rises from 45.8% to 77.1% — a 68.3% relative increase. Mean CVSS severity increases in parallel (6.4 → 7.9), indicating that longer interactions not only produce more violations but more severe ones.

Interaction Length	Violation Rate	Avg CVSS	n	Δ vs. Single-Turn
Single-turn (1)	45.8%	6.4	48	—
3-turn	60.4%	7.1	48	+31.9%
5-turn	70.8%	7.6	48	+54.6%
7-turn	77.1%	7.9	48	+68.3%

Table 3. Violation rate by interaction length (prompt injection + identity spoofing categories, n=48 per condition).

4.3 RQ3: Effect of Tool Access Level

Tool access level has a large, monotonic effect on violation rates (Table 4). Full tool access increases violation rates 109% over no-tool baseline (71.9% vs. 34.4%). This gap cannot be explained by the additional attack surface alone — the nature of failures changes qualitatively: no-tool agents fail primarily through instruction override, while full-tool agents additionally exhibit trust propagation via tool output and data exfiltration through parameter manipulation.

Tool Access	Violation Rate	Avg CVSS	n	Primary New Failure Mode
No tools (chat only)	34.4%	5.8	32	Instruction override
Read-only tools	46.9%	6.7	32	Data exfiltration
Read-write (limited)	59.4%	7.3	32	Privilege escalation via tool call
Full tool suite	71.9%	8.2	32	Trust propagation + all prior

Table 4. Violation rate and severity by tool access level (all attack categories, n=32 per condition).

5. Analysis: Why Agents Fail

Across 190 observed violations, we identify five recurring failure modes (Table 5). These are not independent: a single violation may exhibit multiple patterns, particularly in multi-turn interactions.

Failure Mode	% of Violations	Primary Triggers
Instruction Override	28.4%	Direct injection, identity spoofing
Trust Propagation	22.1%	Indirect injection, identity spoofing
Context Drift	19.5%	Multi-turn interactions, goal hijacking
Scope Ambiguity Exploit	16.8%	Privilege escalation, role confusion
Helpfulness Override	13.2%	Goal hijacking, safety bypass

Table 5. Failure mode taxonomy across 190 observed violations.

Context drift is the most concerning finding for production deployments: agents that correctly refuse a constraint violation in turn 1 will frequently comply by turn 5–7, without any explicit override. This suggests that constraint adherence is not robustly maintained across context window growth — a failure mode invisible to single-turn evaluation.

Trust propagation — where authority granted to one principal (e.g., the system prompt author) is implicitly extended to a downstream impersonator — accounts for 22.1% of violations and explains why indirect injection so significantly outperforms direct injection. Agents appear to treat tool output as carrying system-level trust, a behavior not prevented by any of the tested system prompt formulations.

6. Implications for AI Safety

Our findings have direct implications for how agentic AI systems should be designed, evaluated, and deployed:

Static safeguards fail in dynamic agent workflows. System prompts that successfully constrain behavior in isolation degrade under multi-turn adversarial pressure. Safety evaluation must be conducted at the *workflow level*, not the prompt level.

Alignment must be evaluated across interaction sequences. The 68.3% increase in violation rate from single-turn to 7-turn conditions means that standard red-team benchmarks — which predominantly use single-turn evaluation — substantially underestimate real-world vulnerability.

Tool access is not just an attack surface expansion — it introduces qualitatively new failure modes. No-tool agents fail through text-level manipulation; tool-enabled agents additionally fail through trust propagation in tool outputs and parameter-level data exfiltration. These require distinct defenses.

Trust models must be explicit and persistent. The most common mitigation observed in lower-violation configurations was explicit, detailed identity modeling in the system prompt (e.g., specifying exactly which principals are trusted and what authority they hold). Implicit trust assumptions are consistently exploitable.

7. Limitations

This study has several limitations that scope its conclusions. First, all evaluations were conducted against a single model (Claude Sonnet 4, May 2026); results may differ across model families and versions. Second, the target agent (OrderBot) represents one domain and constraint structure; results may not generalize to agents with different tool sets or authority models. Third, violation judgments are made by the same model family used for probe generation, introducing potential consistency bias. Fourth, $n=24-48$ per condition, while sufficient for the observed effect sizes, limits statistical power for subgroup analysis. Future work should evaluate across multiple models, domains, and with human-in-the-loop violation verification.

8. Conclusion

We present the first systematic empirical study of safety constraint violations in LLM-based agents across attack type, interaction length, and tool access level. Our core finding is that LLM agents are substantially more vulnerable than single-turn evaluation suggests: violation rates reach 77.1% under multi-turn adversarial pressure and 71.9% with full tool access. Current prompt-level alignment techniques do not prevent context drift or trust propagation — the two most common failure modes we observe.

We release agent-redteam as an open evaluation infrastructure to enable reproducible research in this area. We hope this work motivates both stronger evaluation standards for agentic deployments and new alignment techniques that account for the dynamic, multi-step nature of agent interaction.

References

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